

FleetSpark — Bike Financing & Payment Management Module

Version: 1.0

Date: June 11, 2026

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Status: Research-backed specification

1. Executive Summary

The e-boda financing market in Uganda is growing fast but is plagued by **predatory lending practices, high default rates, and zero consumer protection**. Riders are getting trapped in debt cycles with aggressive repossession, hidden fees, and no grace periods. FleetSpark can solve this by providing a **transparent, technology-driven financing management platform** that protects both the lender (fleet operator/financier) and the rider.

2. How E-Boda Financing Works in Uganda (Research Findings)

Current Market Terms

Provider	Bike Price	Down Payment	Weekly Payment	Term	Total Cost	Effective APR
MOGO (Zembo Thunder)	UGX 4,350,000	UGX 600,000	UGX 69,109	24 mo (104 wks)	~UGX 7,779,336	~45%
MOGO (Spiro Ekon)	UGX 4,850,000	UGX 600,000	UGX 57,924	30 mo (130 wks)	~UGX 8,130,120	~42%
MOGO (Spiro Ekon)	UGX 4,850,000	UGX 600,000	UGX 63,545	24 mo (104 wks)	~UGX 7,208,680	~38%
Watu Credit	~UGX 3,500,000	UGX 700,000	UGX 80,000-120,000	18-24 mo	~UGX 8,000,000+	~50%+
Tugende	~UGX 3,000,000	Varies	Varies	18-24 mo	Varies	~40-60%
Spiro Direct	~UGX 4,000,000	UGX 195,000	Varies	24 mo	Varies	Varies

Key Observations

- Down payments range from UGX 195,000 to 700,000** — the "UGX 100,000" Cliff mentioned is likely a promotional or older figure. Current minimum is ~UGX 600,000 for electric bikes.
- Effective interest rates are 38-60% APR** — extremely high, but justified by high default risk and operational costs in informal markets.
- Weekly payment model** dominates — aligns with boda boda daily earning cycles.

4. **Total cost is 1.5x-2.5x the cash price** — riders pay a massive premium for credit.
5. **No government regulation** — Ministry of Trade says it's "service sector," Ministry of Transport says it's not their jurisdiction. Riders have zero consumer protection.

3. What Riders Are Complaining About (Research-Backed)

Top Complaints

#	Complaint	Frequency	Severity
1	Aggressive repossession — bikes taken after 1-2 missed payments, even near end of loan term	Very High	Critical
2	Hidden fees — storage fees after repossession, discharge fees (UGX 97,000+), "processing fees"	High	Critical
3	No grace period — illness, accident, family emergency = no flexibility	Very High	Critical
4	Contract not understood — English-only contracts, riders speak Luganda, no translation	High	High
5	Bike sold after repossession — even if rider was 90% paid off	Medium	Critical
6	Harassment — lenders send agents to stages, publicly shame defaulters	High	High
7	Market saturation — too many bodas, earnings down, can't meet payments	High	Medium
8	Overuse/overwork — riders overwork bikes to meet payments, premature wear	Medium	Medium
9	No insurance — accident or theft = rider still owes full loan	High	Critical
10	Logbook held by lender — rider can't sell or use bike as collateral elsewhere	Medium	Medium

The Defaulter's Story (Typical)

1. Rider misses 1-2 weekly payments (UGX 60,000-120,000)
2. Lender sends agent to boda stage
3. Bike repossessed (often with GPS kill switch or physical seizure)
4. Storage fees accrue (UGX 5,000-10,000/day)
5. Rider given 7-14 days to clear arrears + storage
6. If not paid, bike sold — often for less than outstanding balance
7. Rider still owes the difference (deficiency balance)
8. Rider's guarantor (family member) also liable

4. How Defaulter Recovery Currently Works

Technology Used

Method	How It Works	Used By
GPS Kill Switch	Remote engine disable via telematics. Bike stops when rider reaches home or a swap station.	MOGO, Watu, Tugende
GPS Tracking	Real-time location monitoring. Agents know exactly where the bike is.	All major lenders
Physical Repossession	Agents physically take the bike from the stage or rider's home.	All lenders
Logbook Holding	Lender holds the bike's logbook. Rider can't sell or transfer.	All lenders
Guarantor System	Another person (family member) guarantees the loan. Liable if rider defaults.	MOGO, Watu
Stage-Based Collection	Lenders work with boda stage chairmen to track riders.	All lenders

The GPS Kill Switch Flow

```
Rider misses payment
→ Lender sends SMS warning (Day 1-3)
→ If no payment: Remote GPS kill switch activated
→ Bike stops at next swap station or when ignition turned off
→ Rider can't restart
→ Lender's agent collects bike from known location
→ Bike taken to lender's yard
→ Rider given 7-14 days to clear arrears
→ If not paid: bike sold
```

5. FleetSpark's Solution: Financing Management Module

Core Philosophy

Transparency + Flexibility + Technology = Lower defaults, happier riders, better returns

Module Components

5.1 Loan Contract Management

- Digital loan contracts with **full terms visible** in the rider's language
- **E-signature** or SMS confirmation (for low-literacy riders)
- Contract stored in system, accessible to rider anytime
- Automatic calculation of: total cost, weekly payment, effective interest rate
- **No hidden fees** — all charges shown upfront

5.2 Payment Tracking

- **Mobile Money integration** (MTN MoMo, Airtel Money) — riders pay via USSD
- **Automatic payment detection** — system matches MoMo payments to rider accounts
- **Payment history** — full ledger per rider
- **Receipt generation** — automatic SMS receipt after each payment
- **Payment reminders** — SMS 1 day before due date

5.3 Flexible Repayment Engine

- **Standard schedule** — weekly payments as contracted
- **Early payment bonus** — small discount for paying early (incentive)
- **Partial payment** — accept partial amounts, reduce arrears
- **Payment rescheduling** — allow 1 rescheduling per quarter (manager approval)
- **Grace period** — configurable (e.g., 7 days) before marking as default
- **Emergency pause** — up to 2 weeks pause for documented emergencies (hospital, accident)

5.4 Defaulter Management (Graduated Response)

Stage	Days Overdue	Action	Channel
Reminder	1-3 days	Friendly SMS reminder	SMS
Warning	4-7 days	Formal warning SMS + call	SMS + Phone
Grace offer	8-14 days	Offer payment plan adjustment	Phone call
Final notice	15-21 days	Final warning before GPS lock	SMS + Phone
GPS lock	22-30 days	Remote immobilization	GPS kill switch
Repossession	30+ days	Physical repossession	Field agent
Recovery	Post-repo	Arrears notice, storage fees, sale	Formal letter

Key difference from current market: FleetSpark gives riders **multiple chances** and **clear communication** at every stage. No surprise repossessions.

5.5 GPS Integration for Recovery

- **Real-time tracking** — always know where financed bikes are
- **Geofencing** — alert if bike leaves designated operating area
- **Remote immobilization** — kill switch integration (via Teltonika SECO or similar)
- **Last known location** — if bike is moved without authorization
- **Recovery mode** — when activated, increases location update frequency to every 30 seconds

5.6 Guarantor Management

- Store guarantor details (name, phone, relationship, NIN)
- Automatic SMS to guarantor when rider defaults
- Guarantor payment portal (guarantor can pay on rider's behalf)
- Guarantor liability tracking

5.7 Insurance Integration (Future)

- Partner with insurance companies (APA, Jubilee, UAP)
- Automatic insurance for financed bikes
- Accident/theft = insurance pays off loan
- Rider protected, lender protected

5.8 Reporting & Analytics

- **Portfolio health** — % on-time, % overdue, % defaulted
 - **Default rate by cohort** — which month's loans are defaulting most?
 - **Rider risk score** — based on payment history, age, location
 - **Recovery rate** — % of defaulted bikes recovered
 - **Revenue tracking** — interest earned, fees collected, losses from defaults
-

6. Database Additions for Financing Module

New Tables (to add to existing schema)

```
// — Loan Products —

model LoanProduct {
  id          String      @id @default(uuid())
  orgId       String      @map("org_id")
  name        String      // e.g., "E-Boda Standard 24mo"
  vehicleType VehicleType @map("vehicle_type")
  bikePriceUgx Int         @map("bike_price_ugx")
  minDepositUgx Int        @map("min_deposit_ugx")
  interestRatePct Float    @map("interest_rate_pct") // Annual
  termWeeks   Int          @map("term_weeks")
  weeklyPaymentUgx Int     @map("weekly_payment_ugx")
  totalCostUgx Int         @map("total_cost_ugx")
  isActive    Boolean      @default(true) @map("is_active")
  createdAt   DateTime     @default(now()) @map("created_at")
  updatedAt   DateTime     @updatedAt @map("updated_at")

  organization Organization @relation(fields: [orgId], references: [id], onDelete: Cascade)
  loans        Loan[]

  @@map("loan_products")
}

// — Loans —

model Loan {
  id          String      @id @default(uuid())
  orgId       String      @map("org_id")
  loanProductId String    @map("loan_product_id")
  riderId     String      @map("rider_id") // References drivers table
  vehicleId   String      @map("vehicle_id") // References vehicles table
  guarantorName String?   @map("guarantor_name")
  guarantorPhone String?  @map("guarantor_phone")
  guarantorNin String?    @map("guarantor_nin")
  guarantorRelation String? @map("guarantor_relation")

  // Loan terms
  bikePriceUgx Int         @map("bike_price_ugx")
  depositUgx Int           @map("deposit_ugx")
  amountFinancedUgx Int     @map("amount_financed_ugx")
  interestRatePct Float    @map("interest_rate_pct")
  termWeeks   Int          @map("term_weeks")
  weeklyPaymentUgx Int     @map("weekly_payment_ugx")
  totalRepayableUgx Int    @map("total_repayable_ugx")

  // Status tracking
  status      LoanStatus @default(ACTIVE)
  startDate   DateTime   @map("start_date")
  expectedEndDate DateTime @map("expected_end_date")
  actualEndDate DateTime? @map("actual_end_date")

  // Balance tracking
  totalPaidUgx Int         @default(0) @map("total_paid_ugx")
  totalInterestUgx Int     @default(0) @map("total_interest_ugx")
  totalFeesUgx Int         @default(0) @map("total_fees_ugx")
  arrearsUgx Int           @default(0) @map("arrears_ugx")
  remainingBalance Int     @map("remaining_balance")

  // Default tracking
  weeksOverdue Int         @default(0) @map("weeks_overdue")
  lastPaymentDate DateTime? @map("last_payment_date")
  nextPaymentDue DateTime  @map("next_payment_due")
  defaultDate  DateTime?   @map("default_date")
}
```

```

repossessedAt      DateTime?      @map("repossessed_at")
repossessedBy      String?         @map("repossessed_by")

// GPS immobilization
gpsImmobilized     Boolean         @default(false) @map("gps_immobilized")
gpsImmobilizedAt   DateTime?       @map("gps_immobilized_at")

createdAt          DateTime         @default(now()) @map("created_at")
updatedAt          DateTime         @updatedAt @map("updated_at")

organization        Organization    @relation(fields: [orgId], references: [id], onDelete: Cascade)
loanProduct         LoanProduct    @relation(fields: [loanProductId], references: [id])
rider               Driver          @relation(fields: [riderId], references: [id])
vehicle             Vehicle         @relation(fields: [vehicleId], references: [id])
payments            Payment[]
paymentReschedules  PaymentReschedule[]
defaulterActions    DefaulterAction[]

@@index([orgId, status])
@@index([riderId])
@@index([nextPaymentDue])
@@index([status, weeksOverdue])
@@map("loans")
}

enum LoanStatus {
  PENDING          // Contract signed, waiting for deposit
  ACTIVE           // Payments ongoing
  COMPLETED        // Fully paid
  DEFAULTED        // In default (missed payments)
  REPOSSESSED      // Bike repossessed
  WRITTEN_OFF      // Unrecoverable
  RESTRUCTURED     // Terms renegotiated
}

// — Payments —————

model Payment {
  id              String           @id @default(uuid())
  loanId          String           @map("loan_id")
  amountUgx      Int              @map("amount_ugx")
  paymentMethod   PaymentMethod    @map("payment_method")
  transactionId   String?          @map("transaction_id") // MoMo transaction reference
  transactionRef  String?          @map("transaction_ref") // External reference
  weekNumber      Int              @map("week_number") // Which week of the loan
  dueDate         DateTime         @map("due_date")
  paidAt          DateTime         @map("paid_at")
  status          PaymentStatus    @default(COMPLETED)
  channel         String?          // "MTN_MOMO", "AIRTEL_MONEY", "CASH", "BANK"
  notes           String?
  createdAt       DateTime         @default(now()) @map("created_at")

  loan            Loan             @relation(fields: [loanId], references: [id], onDelete: Cascade)

  @@index([loanId, weekNumber])
  @@index([transactionId])
  @@index([paidAt])
  @@map("payments")
}

enum PaymentMethod {
  MOBILE_MONEY
  CASH
  BANK_TRANSFER

```



```

    GUARANTOR_PAYMENT
    DEDUCTION // From rider's wallet/earnings
}

enum PaymentStatus {
    PENDING
    COMPLETED
    FAILED
    REFUNDED
}

// — Payment Reschedules —————

model PaymentReschedule {
    id          String      @id @default(uuid())
    loanId       String      @map("loan_id")
    oldDueDate   DateTime    @map("old_due_date")
    newDueDate   DateTime    @map("new_due_date")
    reason       String
    approvedBy   String      @map("approved_by") // User ID
    createdAt    DateTime    @default(now()) @map("created_at")

    loan         Loan        @relation(fields: [loanId], references: [id], onDelete: Cascade)

    @@map("payment_reschedules")
}

// — Defaulter Actions —————

model DefaulterAction {
    id          String      @id @default(uuid())
    loanId       String      @map("loan_id")
    actionType   DefaulterActionType @map("action_type")
    daysOverdue  Int         @map("days_overdue")
    message      String?
    sentTo       String?     // Phone number or user ID
    sentBy       String      @map("sent_by") // User ID or "SYSTEM"
    response     String?     // Rider's response
    actionDate   DateTime    @default(now()) @map("action_date")

    loan         Loan        @relation(fields: [loanId], references: [id], onDelete: Cascade)

    @@index([loanId, actionDate])
    @@map("defaulter_actions")
}

enum DefaulterActionType {
    SMS_REMINDER
    SMS_WARNING
    PHONE_CALL
    GRACE_OFFER
    FINAL_NOTICE
    GPS_LOCK
    GPS_UNLOCK
    REPOSSESSION_ORDER
    REPOSSESSION_COMPLETE
    GUARANTOR_NOTICE
    PAYMENT_RECEIVED
    RESCHEDULE_OFFER
    RESCHEDULE_APPROVED
    INSURANCE_CLAIM
    LEGAL_NOTICE
}

```

```
// — Rider Wallets (for in-app earnings tracking) —

model RiderWallet {
  id          String      @id @default(uuid())
  riderId     String      @unique @map("rider_id")
  loanId      String      @map("loan_id")
  balanceUgx  Int          @default(0) @map("balance_ugx")
  totalEarnedUgx Int      @default(0) @map("total_earned_ugx")
  totalPaidUgx Int        @default(0) @map("total_paid_ugx")
  totalDeductionsUgx Int  @default(0) @map("total_deductions_ugx")
  lastActivityAt DateTime? @map("last_activity_at")
  createdAt   DateTime     @default(now()) @map("created_at")
  updatedAt   DateTime     @updatedAt @map("updated_at")

  rider       Driver       @relation(fields: [riderId], references: [id], onDelete: Cascade)
  loan        Loan         @relation(fields: [loanId], references: [id])
  transactions WalletTransaction[]

  @@map("rider_wallets")
}

model WalletTransaction {
  id          String      @id @default(uuid())
  walletId     String      @map("wallet_id")
  amountUgx    Int          @map("amount_ugx")
  type         WalletTransactionType
  description   String?
  reference     String?     // External reference
  createdAt     DateTime     @default(now()) @map("created_at")

  wallet       RiderWallet @relation(fields: [walletId], references: [id], onDelete: Cascade)

  @@index([walletId, createdAt])
  @@map("wallet_transactions")
}

enum WalletTransactionType {
  EARNING      // Daily earning added
  LOAN_DEDUCTION // Weekly payment deducted
  BONUS        // Early payment bonus
  PENALTY      // Late payment fee
  WITHDRAWAL   // Rider cashes out
  ADJUSTMENT   // Manual adjustment
}

```

7. API Endpoints for Financing Module

Loans

Method	Endpoint	Description
POST	/api/v1/loans	Create a new loan
GET	/api/v1/loans	List loans (filter by status, rider, overdue)
GET	/api/v1/loans/:id	Loan detail with payment history
PUT	/api/v1/loans/:id	Update loan terms
POST	/api/v1/loans/:id/reschedule	Reschedule payment
POST	/api/v1/loans/:id/immobilize	GPS lock the bike
POST	/api/v1/loans/:id/unimmobilize	GPS unlock the bike
POST	/api/v1/loans/:id/reposess	Mark as reposessed
POST	/api/v1/loans/:id/complete	Mark as fully paid

Payments

Method	Endpoint	Description
POST	/api/v1/payments	Record a payment
GET	/api/v1/payments	List payments (filter by loan, date)
GET	/api/v1/payments/upcoming	Payments due in next 7 days
GET	/api/v1/payments/overdue	Overdue payments
POST	/api/v1/payments/webhook/momo	MoMo payment webhook

Defaulter Management

Method	Endpoint	Description
GET	/api/v1/defaulters	List all defaulters
POST	/api/v1/defaulters/:loanId/action	Log a defaulter action
GET	/api/v1/defaulters/:loanId/actions	Action history for a loan
POST	/api/v1/defaulters/bulk-sms	Send bulk SMS to defaulters
GET	/api/v1/defaulters/analytics	Default rate, recovery rate, etc.

Rider Wallets

Method	Endpoint	Description
GET	/api/v1/wallets/:riderId	Wallet balance + transactions
POST	/api/v1/wallets/:riderId/credit	Add earning
POST	/api/v1/wallets/:riderId/debit	Deduct payment
POST	/api/v1/wallets/:riderId/withdraw	Rider cash-out

8. Background Jobs

Job	Frequency	Description
payment-due-reminder	Daily 8AM EAT	SMS to riders with payments due in 2 days
overdue-check	Daily 9AM EAT	Identify overdue loans, update arrears
defaulter-escalation	Daily 10AM EAT	Run graduated response for defaulters
gps-lock-check	Every 6 hours	Auto-lock bikes past GPS lock threshold
portfolio-report	Weekly (Monday)	Generate portfolio health report
interest-accrual	Daily midnight	Calculate and add interest to balances
wallet-deduction	Daily 11PM EAT	Auto-deduct weekly payment from rider wallet

9. Key Metrics to Track

Metric	Formula	Target
On-time payment rate	On-time payments / Total due	>80%
Default rate	Defaults / Total loans	<15%
Recovery rate	Recovered bikes / Defaulted bikes	>90%
Average days to recovery	Avg days from default to recovery	<21 days
Portfolio at risk	Value of loans >30 days overdue / Total portfolio	<10%
Rider satisfaction	Survey score	>4/5
Effective yield	Interest earned / Average portfolio	>30% APR
Loss rate	Written-off value / Total portfolio	<5%

10. Competitive Advantage

Current lenders (MOGO, Watu, Tugende) use:

- Basic spreadsheets or simple databases
- Manual payment tracking (MoMo codes)
- Aggressive, one-size-fits-all defaulter treatment
- No rider-facing portal

FleetSpark provides:

- **Full loan lifecycle management** — from contract to completion
- **Automated payment matching** — MoMo webhook integration
- **Graduated defaulter response** — configurable, fair, documented
- **Rider self-service portal** — riders can see their balance, payment history, request rescheduling
- **GPS integration** — automated immobilization with audit trail
- **Portfolio analytics** — data-driven lending decisions

11. Risks & Mitigations

Risk	Likelihood	Mitigation
Riders can't afford payments	High	Flexible terms, partial payments, rescheduling
GPS kill switch doesn't work	Medium	Multiple kill methods (remote + at swap station)
MoMo payment matching fails	Medium	Manual reconciliation + rider self-report
Guarantor disputes	Medium	Clear contract, SMS documentation
Regulatory changes	Low	Build compliant from day one, engage Ministry of Trade
Fraud (fake identities)	Medium	NIN verification, stage chairman confirmation
Bike theft	Medium	GPS tracking + insurance integration

12. Pricing for Financing Module

Feature	Price
Loan management (up to 50 loans)	Included in base plan
50-200 loans	+\$100/mo
200-500 loans	+\$250/mo
500+ loans	Custom
GPS immobilization integration	+\$50/mo
MoMo auto-matching	+\$50/mo
Insurance integration	Revenue share with insurer

Sources: MOGO Uganda (mogo.co.ug), Global Press Journal (2022), PC Tech Magazine (2024), Daily Monitor, Africa-Press, rider complaints via Facebook/TikTok/Instagram